

Hands-on Lab: Generative AI for Data Generation and Augmentation

Estimated time needed: 30 minutes

Introduction

One of the principal advantages of generative AI is its ability to generate realistic synthetic data. Synthetic data is generated when a pretrained generative model responds to a prompt, creates new data samples, or performs transfer learning on a given dataset. In addition, it generates samples that augment the existing data set while preserving its statistical distribution and interpretability.

In this lab, you will learn how to use generative AI to generate synthetic data samples and transfer learning on a given dataset.

Learning objectives

By the end of this lab, you will be able to understand and apply key concepts of synthetic data generation using generative AI tools in practical scenarios.

- Understand the concept of synthetic data generation using generative AI
- Use tools such as **Tonic Fabricate** and **ilovecsv** to create synthetic data samples

- Explore how generative AI can enhance datasets for analysis, modeling, and other downstream tasks

Insurance dataset overview

The dataset used in this lab includes insurance records.

This dataset is a cleaned version of the original dataset, meaning the data has been processed to remove errors, handle missing values, and ensure consistency.

The original dataset, [Medical Insurance Price Prediction](#), contains raw, unprocessed data and is available on [Kaggle](#) under the [CC0 1.0 Universal License](#), which allows free use and distribution.

You can download the cleaned dataset file directly from the link provided here: [Insurance Dataset](#)

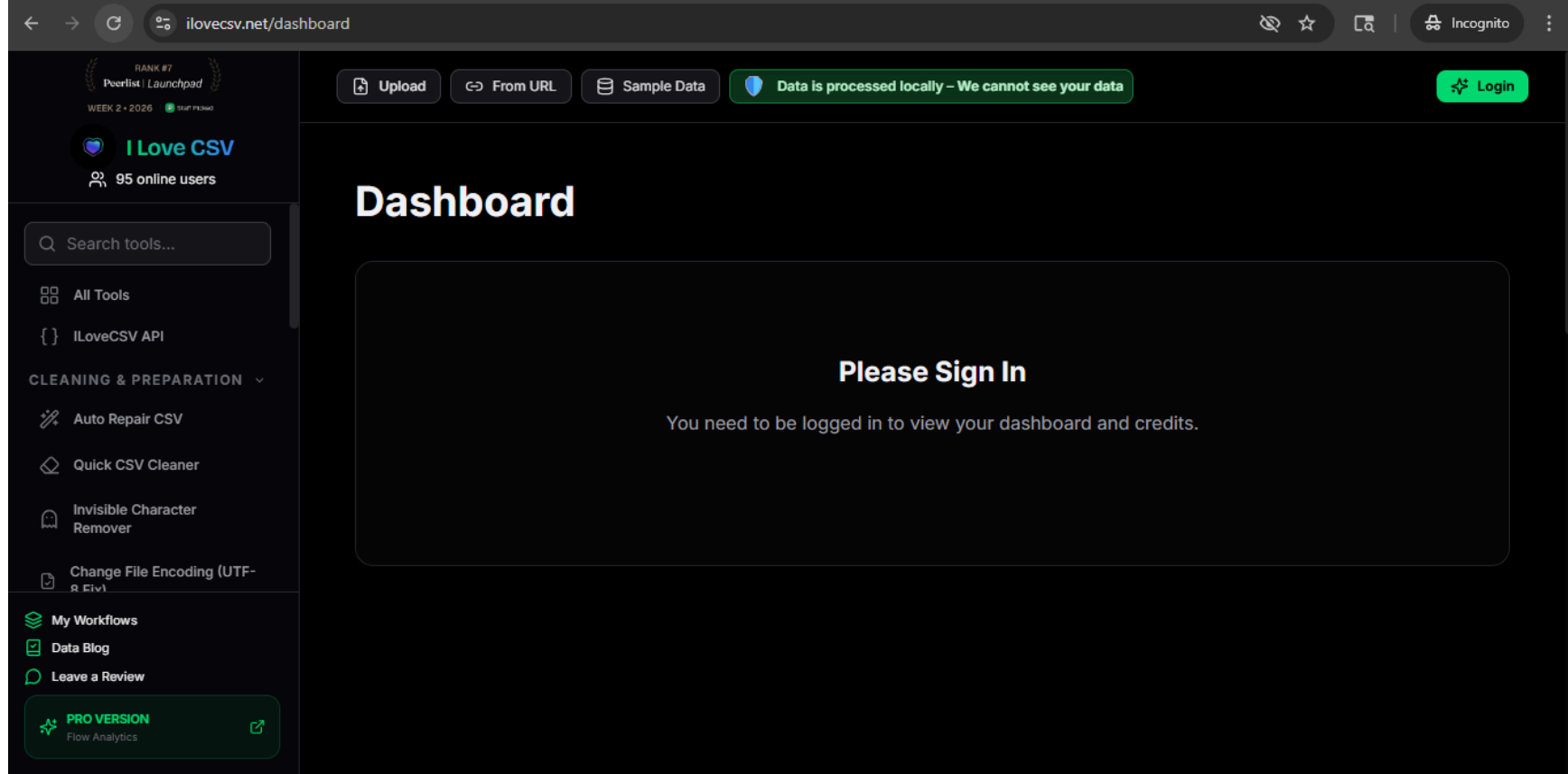
Option 1: ilovecsv Tool

The **ilovecsv** tool is a simple, GUI-based tool. You will now explore how to use it.

1. Open the website

Click this [ilovecsv](#) link to open the website and interface.

This link opens in a new browser tab, and you should see a web page that looks similar to the following screen capture:



2. Create an account

You can create an account on this website free of charge, or you can simply log in using your Gmail ID. After you log in, you'll see the following interface:

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I Love CSV
117 online users

Search tools...

All Tools
ILoveCSV API

CLEANING & PREPARATION

Auto Repair CSV
Quick CSV Cleaner
Invisible Character Remover

My Workflows
Dashboard
Data Blog
Leave a Review

PRO VERSION
Flow Analytics

Upload From URL Sample Data

Data is processed locally – We cannot see your data

User Welcome, poojap@skillup.online
0 credits

Online CSV Tools for CSV Lovers

Analyze, clean, and convert your CSV & Excel data online. No installation, privacy-first. No sign-up required.

No sign-up required. Free for files up to 10kb.

Upload File

Search for a tool, e.g., 'Merge' or 'Forecast'...

Cleaning & Preparation

- Auto Repair CSV
- Quick CSV Cleaner
- Invisible Character Remover
- Change File Encoding (UTF-8, UTF-16)
- Merge CSV Files

3. Upload the dataset

- Click on the **Upload** button given at the top of the page, and upload the CSV file of the data set to the interface.

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Start Project

I Love CSV

117 online users

Q Search tools...

All Tools

{ } I LoveCSV API

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Cleaning & Preparation

✂

Auto Repair CSV

🗑

Quick CSV Cleaner

👤

Invisible Character

📄

Change File Encoding (UTF-8)

🔗

Merge CSV Files

- After uploading the dataset, you will be able to see the **insurance_dataset** file listed in the uploaded files section.

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1/20 insurance_dataset.csv Undo Download

User Welcome, poojap@skillup.online
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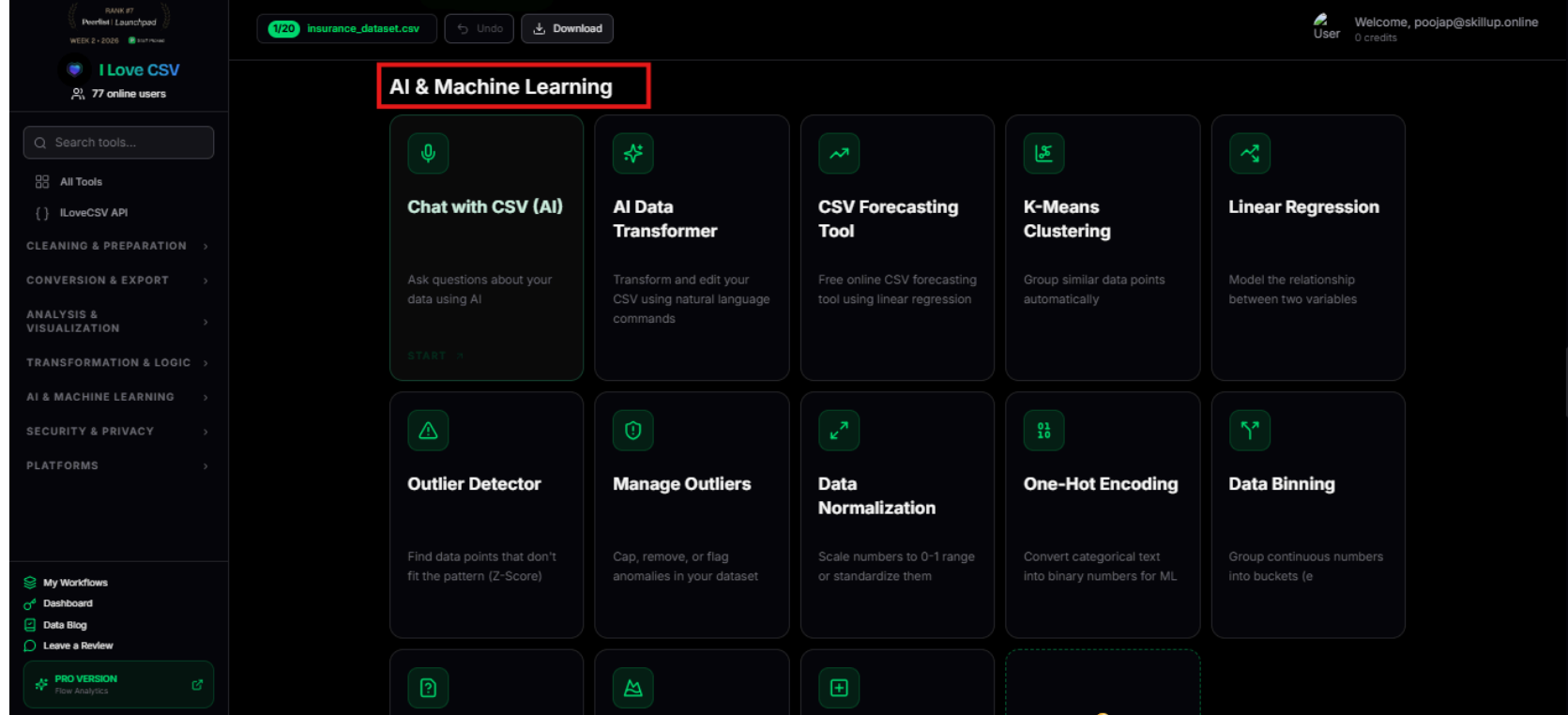
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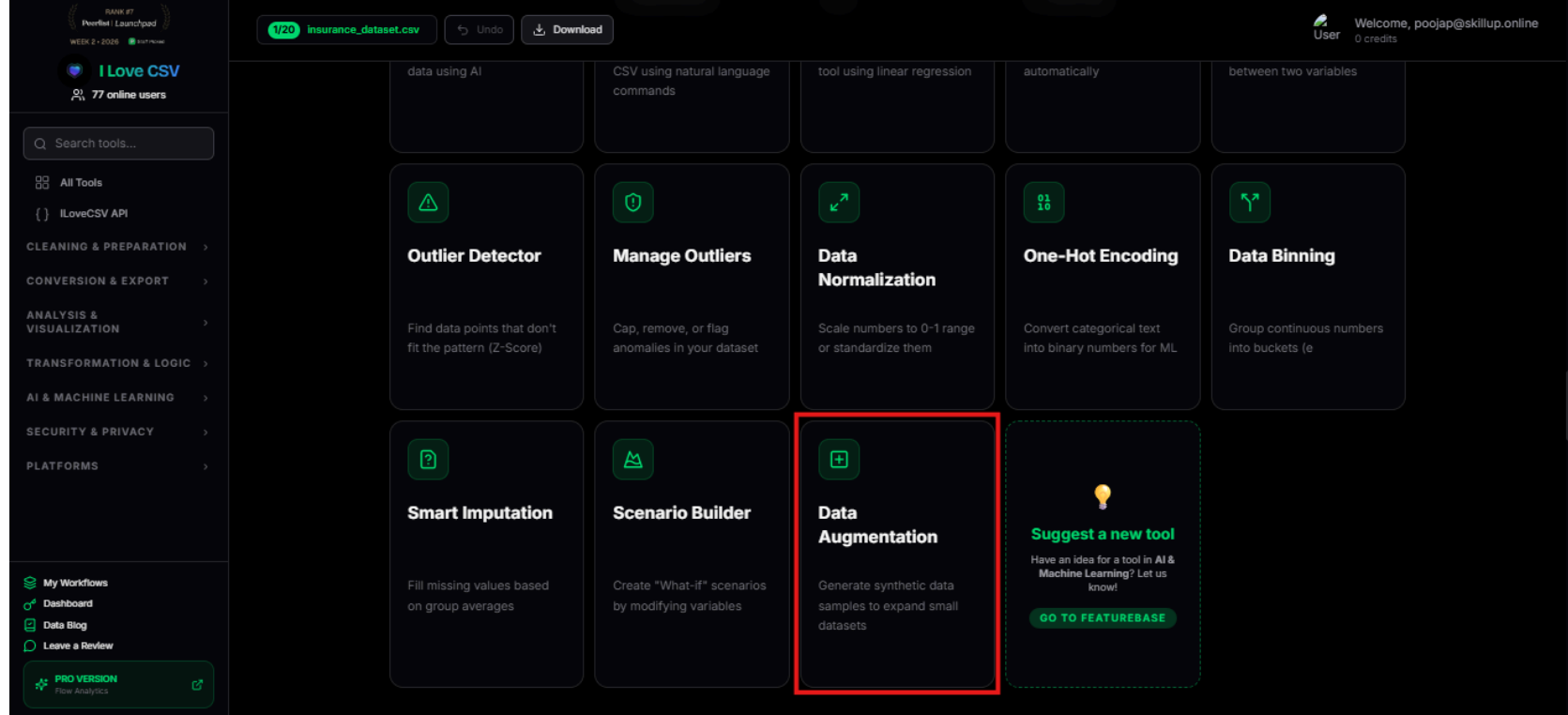
Auto Repair CSV
Quick CSV Cleaner
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Change File Encoding
Merge CSV Files

4. Access the Data Augmentation tool

- You are on the main dashboard page. Scroll down through the available sections until you locate the **AI and Machine Learning** section, or use the left-hand navigation panel to quickly navigate to it directly.



- Within the AI and Machine Learning section, locate the **Data Augmentation Tool** from the available options and click on it to open the tool interface.



5. Augment the data

Statistical augmentation in the ilovecsv tool involves generating synthetic rows from your existing dataset while preserving its statistical properties.

You can use the **Augmentation factor** to adjust the **density (number of rows)** of the synthetic dataset as required.

The screenshot displays the I Love CSV web application. The top header includes a file name '1/20 insurance_dataset.csv', 'Undo' and 'Download' buttons, a user profile 'User', and a welcome message 'Welcome, poojap@skillup.online' with '0 credits'. The left sidebar contains a search bar, a list of tools under 'CLEANING & PREPARATION', and a 'My Workflows' section. The main content area is titled 'Statistical Augmentation' with the subtitle 'Generate high-fidelity synthetic rows based on source distributions.' It features a slider for the 'AUGMENTATION FACTOR' set to '0.5x', with 'LOW DENSITY' and 'HIGH INTENSITY' labels. Below the slider is a text box stating: 'Synthetic records are calculated using Box-Muller transformations for numeric fields, preserving mean and standard deviation.' A large green 'Execute Augmentation' button is positioned below the text box. At the bottom, there are links for 'About' and 'Augment'.

Once you have adjusted the augmentation factor (**for example, 1x**), click the **Execute Augmentation** button.

The screenshot displays the I Love CSV web application interface. On the left is a dark sidebar with a search bar, navigation links for 'All Tools', 'I Love CSV API', and a 'CLEANING & PREPARATION' menu containing options like 'Auto Repair CSV', 'Quick CSV Cleaner', 'Invisible Character Remover', 'Change File Encoding (UTF-8 Fix)', 'Merge CSV Files', 'Advanced CSV Filter', and 'Trim & Sanitize'. Below these are 'My Workflows' links and a 'PRO VERSION' badge. The main content area has a top header with a file upload status '1/20 insurance_dataset.csv', 'Undo', and 'Download' buttons. The user profile 'User' is shown with 'Welcome, poojap@skillup.online' and '0 credits'. The breadcrumb trail is 'Blog > AI & Machine Learning > Augment'. The central section is titled 'Statistical Augmentation' with the subtitle 'Generate high-fidelity synthetic rows based on source distributions.' It features a slider for 'AUGMENTATION FACTOR' set to '1.0x', ranging from 'LOW DENSITY' to 'HIGH INTENSITY'. A text box below the slider states: 'Synthetic records are calculated using Box-Muller transformations for numeric fields, preserving mean and standard deviation.' A large red rectangular box highlights the 'Execute Augmentation' button. At the bottom, there are links for 'About' and 'Augment'.

6. Download the synthetic data

After processing completes, you will see a notification that the operation is complete, similar to what you see in the following screen capture:

The screenshot displays the I Love CSV web application interface. On the left is a dark sidebar with a search bar, navigation links (All Tools, I Love CSV API, CLEANING & PREPARATION, My Workflows, Dashboard, Data Blog, Leave a Review), and a PRO VERSION badge. The top header shows a file named 'insurance_dataset.csv' with 2/20 rows, and buttons for Undo and Download. The user is logged in as 'User' with 0 credits. The main content area is titled 'Statistical Augmentation' with a subtitle 'Generate high-fidelity synthetic rows based on source distributions.' A green notification bar at the top of the main area states 'Operation complete! See Results'. Below this is a slider for 'AUGMENTATION FACTOR' set to '1.0x', with 'LOW DENSITY' and 'HIGH INTENSITY' labels. A text box contains the note: '*Synthetic records are calculated using Box-Muller transformations for numeric fields, preserving mean and standard deviation.*' A large green button labeled 'Execute Augmentation' is positioned below the text box. At the bottom, there are links for 'About' and 'Augment'.

Click the [See Results](#) hyperlink to open and view the final results in the editor.

Dataset Editor

×

Show 10 entries

Search:

	AGE	GENDER	BMI	CHILDREN	SMOKER	REGION	EXPENSES
1	19	female	27.9	0	yes	southwest	16884.92
2	18	male	33.8	1	no	southeast	1725.55
3	28	male	33	3	no	southeast	4449.46
4	33	male	22.7	0	no	northwest	21984.47
5	32	male	28.9	0	no	northwest	3866.86
6	31	female	25.7	0	no	southeast	3756.62
7	46	female	33.4	1	no	southeast	8240.59
8	37	female	27.7	3	no	northwest	7281.51
9	37	male	29.8	2	no	northeast	6406.41
10	60	female	25.8	0	no	northwest	28923.14

Showing page 1 of 268 entries

1

2

3

4

5

6

7

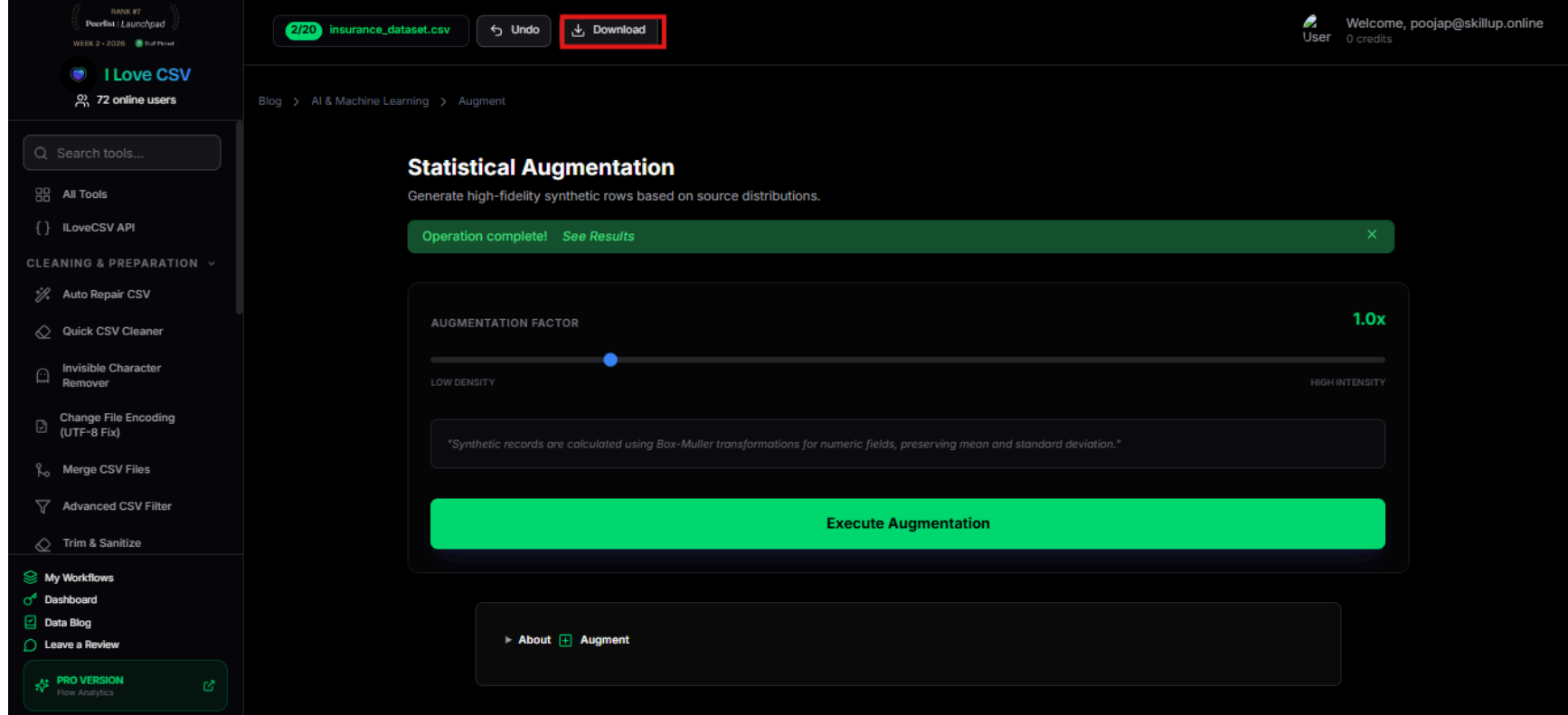
8

9

10

>

Close the results window and return to the previous page. Then click [Download](#) to save the output file to your local machine.



In the final result, you will get 2,676 rows, which is double the original file since you selected a 1× augmentation factor. You can now use this synthetic data set for data science operations, or you can also augment the original data set with these samples.

Option 2: Tonic Fabricate

1. Open the website

Click this [Tonic Fabricate](#) link to open the website and interface.

This link opens in a new browser tab, and you should see a web page that looks similar to the following screen capture:



TONIC FABRICATE

Generate a fake database in seconds.

 [How it works](#)

Create My Account



or use your email for registration

Name

Email

Password

6 characters minimum

Signing up signifies that you are authorized to enter into an agreement with Tonic, and that you have read and agree to our [Terms of Use](#) and [Privacy Policy](#), which will govern all uses of any Tonic product, unless there is a signed agreement between you and Tonic that you agree governs this use.

2. Create an account

You can create an account on this website free of charge, or you can simply log in using your Gmail ID. After you log in, you'll see the following interface:

What data can I fabricate for you today?

Drop in a schema file or describe the type of data you want to get started.

∞ Agent ▾



 Generate data for an online store

 Simulate bank transactions

 Create social media data

3. Data Augmentation

This is a prompt-based tool that lets you customize synthetic data to meet your data science operational requirements. Upload the Insurance dataset using the **Attach a File** option and copy and paste the below prompt in the message box.



plaintext



Generate a synthetic dataset based on the uploaded insurance data file. Preserve the original schema, column

Note: Synthetic data generation in this tool may take some time, so please be patient during the process.

What data can I fabricate for you today?

relationships between variables (e.g., higher age may correlate with higher premiums, and claim amounts should align with policy coverage). Avoid duplicates and unrealistic values. Generate clean, high-quality data suitable for analysis, keeping the synthetic data statistically consistent with the source dataset with approximately 80% similarity/accuracy. Create a dataset that is 1× the size of the original while introducing meaningful variations.

 insurance_dataset.csv (50.2 KB) 

∞ Agent ▾



 Generate data for an online store

 Simulate bank transactions

 Create social media data

Click **Send** to submit the prompt to initiate the process.

TONIC FABRICATE

Default

insurance

insurance_source

PLANS

WORKFLOWS

API

insurance_source

1338 records

Default

SQL

Memory

Import

Export

id	age	gender	bmi	children	smoker	region	expenses
INTEGER	INTEGER	TEXT	REAL	INTEGER	TEXT	TEXT	REAL
1	19	female	27.9	0	yes	southwest	16884.9
2	18	male	33.8	1	no	southeast	1725.55
3	28	male	33	3	no	southeast	4449.46
4	33	male	22.7	0	no	northwest	21984.4
5	32	male	28.9	0	no	northwest	3866.86
6	31	female	25.7	0	no	southeast	3756.62
7	46	female	33.4	1	no	southeast	8240.59
8	37	female	27.7	3	no	northwest	7281.51
9	37	male	29.8	2	no	northeast	6406.41
10	60	female	25.8	0	no	northwest	28923.1
11	25	male	26.2	0	no	northeast	2721.32
12	62	female	26.3	0	yes	southeast	27808.7
13	23	male	34.4	0	no	southwest	1826.84
14	56	female	39.8	0	no	southeast	11090.7
15	27	male	42.1	0	yes	southeast	39611.7

Children

0-5, heavily skewed toward 0 (43%)

Smoker

~80% no, ~20% yes

Region

~24-27% per region

Expenses

Strongly correlated with smoker status + age; smokers avg ~\$32K, non-smokers avg ~\$8.4K

Now I'll create the target table and generate 1,338 synthetic rows preserving these distributions and correlations:

Create synthetic insurance table

6s

Generate synthetic insurance data

24s

All 1,338 synthetic rows generated. Let me now validate the quality by comparing distributions between source and synthetic data:

Ran query source vs synthetic comparison

Ran query region distribution comparison

Ran query expense patterns by age & smoker

Working...

Agent

4. Download the synthetic data

Once the process is complete, you will be prompted with the message: **The data is ready for export or analysis. Would you like me to download it as a CSV?** Type **Yes** to proceed with downloading the dataset.

Default

insurance

PLANS

WORKFLOWS

API

insurance

1338 records

SQL

Memory

Import

Export

id	age	gender	bmi	children	smoker	region	expenses
INTEGER	INTEGER	TEXT	REAL	INTEGER	TEXT	TEXT	REAL
1	33	male	29.2	3	no	southeast	5275.43
2	42	male	32.9	3	no	southeast	16588.7
3	43	female	38.6	2	no	southwest	4876.97
4	38	female	24.7	1	no	southwest	2868.68
5	62	female	28.3	1	no	northeast	14427.8
6	52	male	34	0	no	southwest	2065.15
7	63	female	39.9	0	no	southwest	23015.9
8	44	male	35.6	1	no	southwest	15470.2
9	53	male	35.7	0	yes	southeast	34589.7
10	24	male	26.2	3	no	northeast	4704.95
11	56	female	33	0	no	southeast	16720.5
12	24	female	27.8	1	no	southeast	1970.8
13	40	male	31.5	0	no	northeast	7378.22
14	47	male	33	0	yes	southwest	17645.3
15	49	male	31.2	1	no	southwest	11600.3

Previous

Page 1 of 14

Next

Synthetic Insurance Data Generator

Yes

synthetic_insurance_dataset.csv

text/csv • 49.09 KB

Download

Here's your synthetic insurance dataset as a CSV (49 KB, 1,338 rows). The `id` column has been excluded to keep the format consistent with your original file.

What would you like to do next?

Agent

Click on **Download** to save the data to your local machine.

5. Analyze the report

This tool allows you to view details about the model used to generate the synthetic data.

Copy and paste the below prompt to see the report

<>

plaintext

Provide a detailed report on the model used for generating the synthetic data. Include information about the

After you give the prompt, you will see a detailed report along with visualizations.

Default

insurance

PLANS

WORKFLOWS

API

insurance 1338 records

SQL

Memory

Import

Export

id	age	gender	bmi	children	smoker	region	expenses
INTEGER	INTEGER	TEXT	REAL	INTEGER	TEXT	TEXT	REAL
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Previous

Page 1 of 14

Next

Synthetic Insurance Data Generator

60-64	No	\$15,233	\$16,094	5.7%
60-64	Yes	\$40,631	\$42,178	3.8%

The **monotonic increase** in expenses with age is preserved for both smoker groups. Smokers consistently cost ~3-4x more than non-smokers across all age buckets.

8. Summary Scorecard

Evaluation Dimension	Weight	Score
Numeric Distribution (KS Tests)	30%	96.8%
Categorical Distribution (TVD)	20%	99.0%
Correlation Preservation	25%	96.3%
Discriminator AUC	25%	98.4%
Overall Similarity Score	100%	97.5%

The synthetic dataset exceeds the 80% similarity target across all dimensions, scoring 97.5% overall. The data is statistically consistent with the source, preserves all key relationships, and is indistinguishable from the original by a machine learning classifier.

What would you like to do next?

Agent

Conclusion

Congratulations! You have completed the lab on data augmentation using iLoveCSV and Tonic Fabricate tools.

Author(s)

Pooja Patel



Hands-on Lab: Generative AI for Data Generation and Augmentation

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By the end of this lab, you will be able to understand and apply key concepts of synthetic data generation using generative AI tools in practical scenarios.

- Understand the concept of synthetic data generation using generative AI

- Use tools such as **Tonic Fabricate** and **ilovecsv** to create synthetic data samples
- Augment an existing CSV dataset while maintaining its statistical characteristics
- Explore how generative AI can enhance datasets for analysis, modeling, and other downstream tasks

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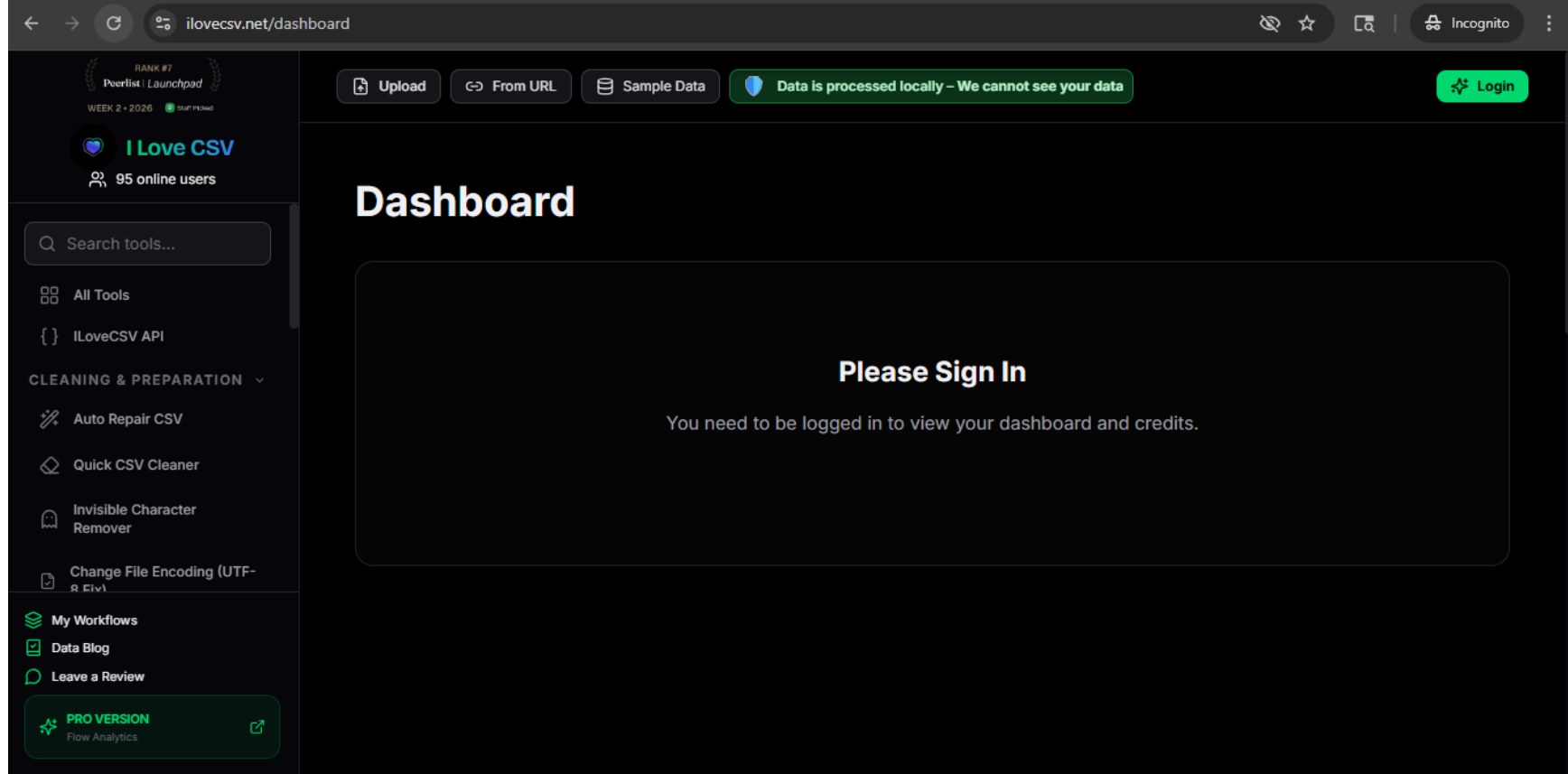
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The screenshot shows the I Love CSV website. The top navigation bar includes buttons for 'Upload', 'From URL', and 'Sample Data', along with a status message 'Data is processed locally - We cannot see your data'. The user profile shows 'User' with '0 credits'. The left sidebar contains a search bar, navigation links like 'All Tools', 'ILoveCSV API', and a 'CLEANING & PREPARATION' section with sub-links for 'Auto Repair CSV', 'Quick CSV Cleaner', and 'Invisible Character Remover'. The main content area features the title 'Online CSV Tools for CSV Lovers', a description of the tools, a 'No sign-up required' banner, a search bar for tools, and a 'Cleaning & Preparation' section with five tool cards: 'Auto Repair CSV', 'Quick CSV Cleaner', 'Invisible Character Remover', 'Change File Encoding', and 'Merge CSV Files'.

3. Upload the dataset

- Click on the **Upload** button given at the top of the page, and upload the CSV file of the data set to the interface.

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1/20 insurance_dataset.csv Undo Download

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0 credits

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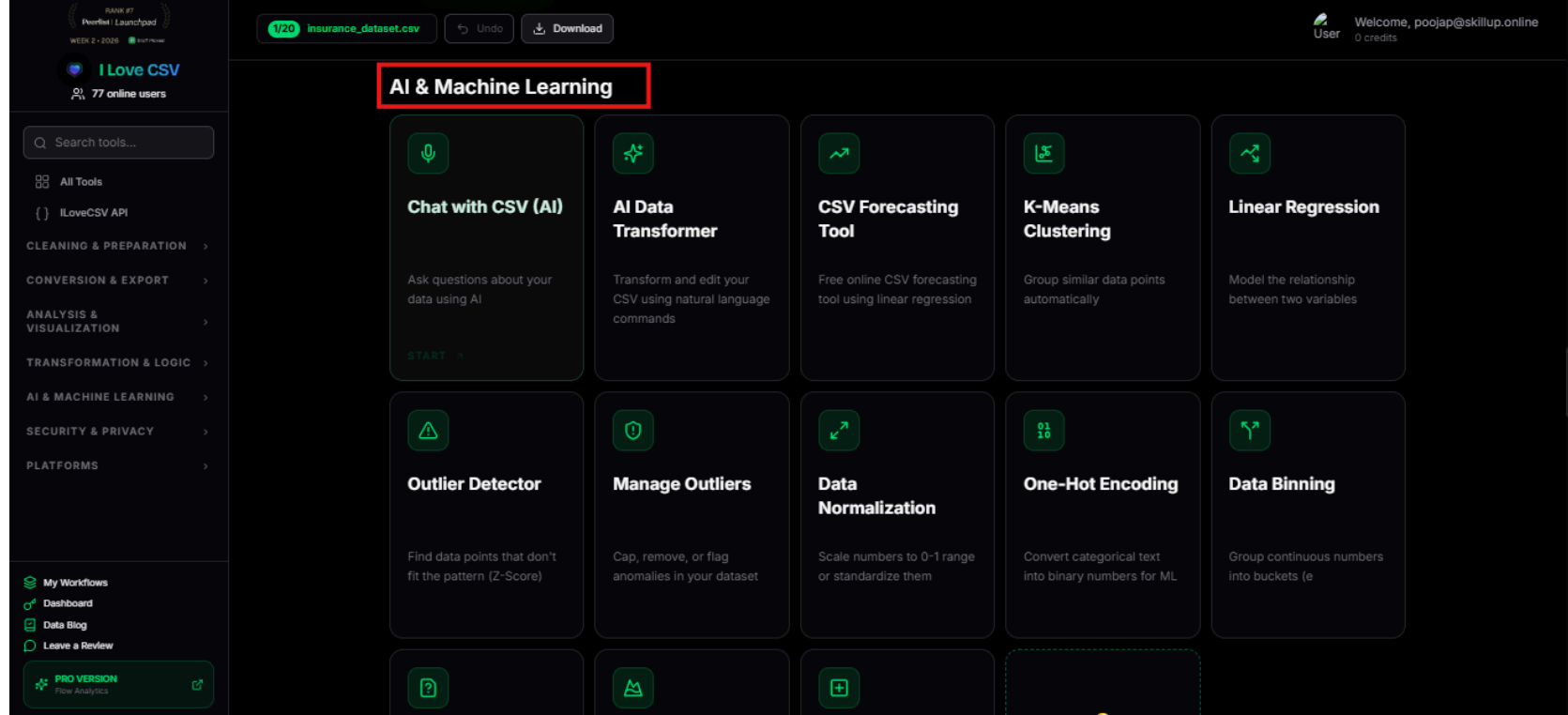
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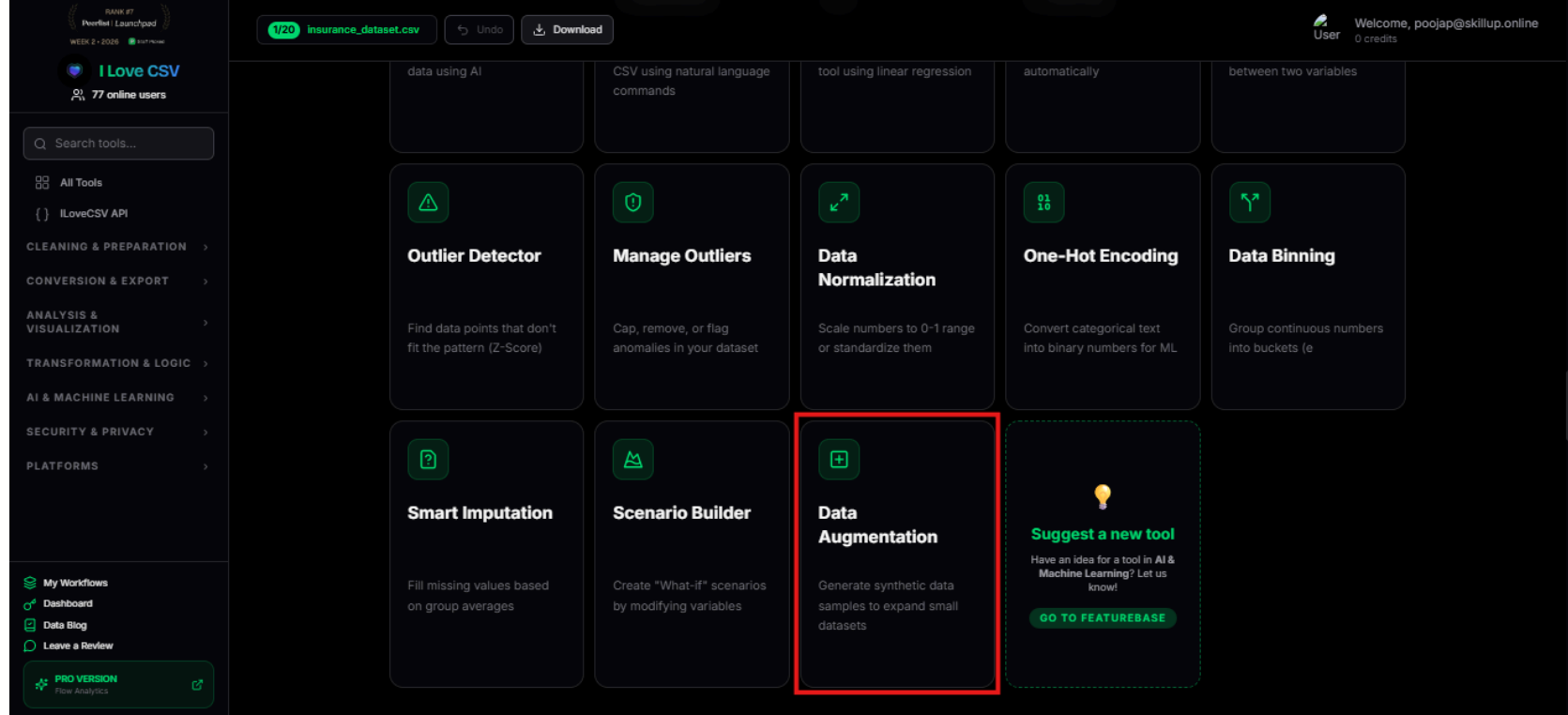
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You can use the **Augmentation factor** to adjust the **density (number of rows)** of the synthetic dataset as required.

The screenshot displays the I Love CSV web application interface. On the left is a dark sidebar with a search bar, a list of tools under 'CLEANING & PREPARATION', and a 'My Workflows' section. The main content area has a top header with a file name '1/20 insurance_dataset.csv', 'Undo', and 'Download' buttons. Below this is a breadcrumb trail: 'Blog > AI & Machine Learning > Augment'. The main heading is 'Statistical Augmentation' with a subtitle 'Generate high-fidelity synthetic rows based on source distributions.' A slider control for 'AUGMENTATION FACTOR' is set to '0.5x', with 'LOW DENSITY' on the left and 'HIGH INTENSITY' on the right. A text box below the slider contains the note: 'Synthetic records are calculated using Box-Muller transformations for numeric fields, preserving mean and standard deviation.' A large orange button labeled 'Execute Augmentation' is positioned below the text box. At the bottom, a dark box contains links for 'About' and 'Augment'.

Once you have adjusted the augmentation factor (**for example, 1x**), click the **Execute Augmentation** button.

The screenshot shows the I Love CSV website interface. At the top, there's a header with a logo, a file name '1/20 insurance_dataset.csv', and buttons for 'Undo' and 'Download'. A user profile 'User' with '0 credits' is visible. The left sidebar contains a search bar and a list of tools under 'CLEANING & PREPARATION', including 'Auto Repair CSV', 'Quick CSV Cleaner', 'Invisible Character Remover', 'Change File Encoding (UTF-8 Fix)', 'Merge CSV Files', 'Advanced CSV Filter', and 'Trim & Sanitize'. The main area is titled 'Statistical Augmentation' with the subtitle 'Generate high-fidelity synthetic rows based on source distributions.' It features a slider for 'AUGMENTATION FACTOR' set to '1.0x', with 'LOW DENSITY' and 'HIGH INTENSITY' labels. Below the slider is a text box containing the note: 'Synthetic records are calculated using Box-Muller transformations for numeric fields, preserving mean and standard deviation.' A large red rectangular button labeled 'Execute Augmentation' is highlighted. At the bottom, there are links for 'About' and 'Augment'.

6. Download the synthetic data

After processing completes, you will see a notification that the operation is complete, similar to what you see in the following screen capture:

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Full Point

I Love CSV

43 online users

Search tools...

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2/20 insurance_dataset.csv

Undo

Download

User

Welcome, poojap@skillup.online

0 credits

Blog > AI & Machine Learning > Augment

Statistical Augmentation

Generate high-fidelity synthetic rows based on source distributions.

Operation complete! See Results

AUGMENTATION FACTOR

1.0x

LOW DENSITYHIGH INTENSITY

"Synthetic records are calculated using Box-Muller transformations for numeric fields, preserving mean and standard deviation."

Execute Augmentation

About

Augment

Click the **See Results** hyperlink to open and view the final results in the editor.

Dataset Editor

Show 10 entries

Search:

	AGE	GENDER	BMI	CHILDREN	SMOKER	REGION	EXPENSES
1	19	female	27.9	0	yes	southwest	16884.92
2	18	male	33.8	1	no	southeast	1725.55
3	28	male	33	3	no	southeast	4449.46
4	33	male	22.7	0	no	northwest	21984.47
5	32	male	28.9	0	no	northwest	3866.86
6	31	female	25.7	0	no	southeast	3756.62
7	46	female	33.4	1	no	southeast	8240.59
8	37	female	27.7	3	no	northwest	7281.51
9	37	male	29.8	2	no	northeast	6406.41
10	60	female	25.8	0	no	northwest	28923.14

Showing page 1 of 268 entries

1

2

3

4

5

6

7

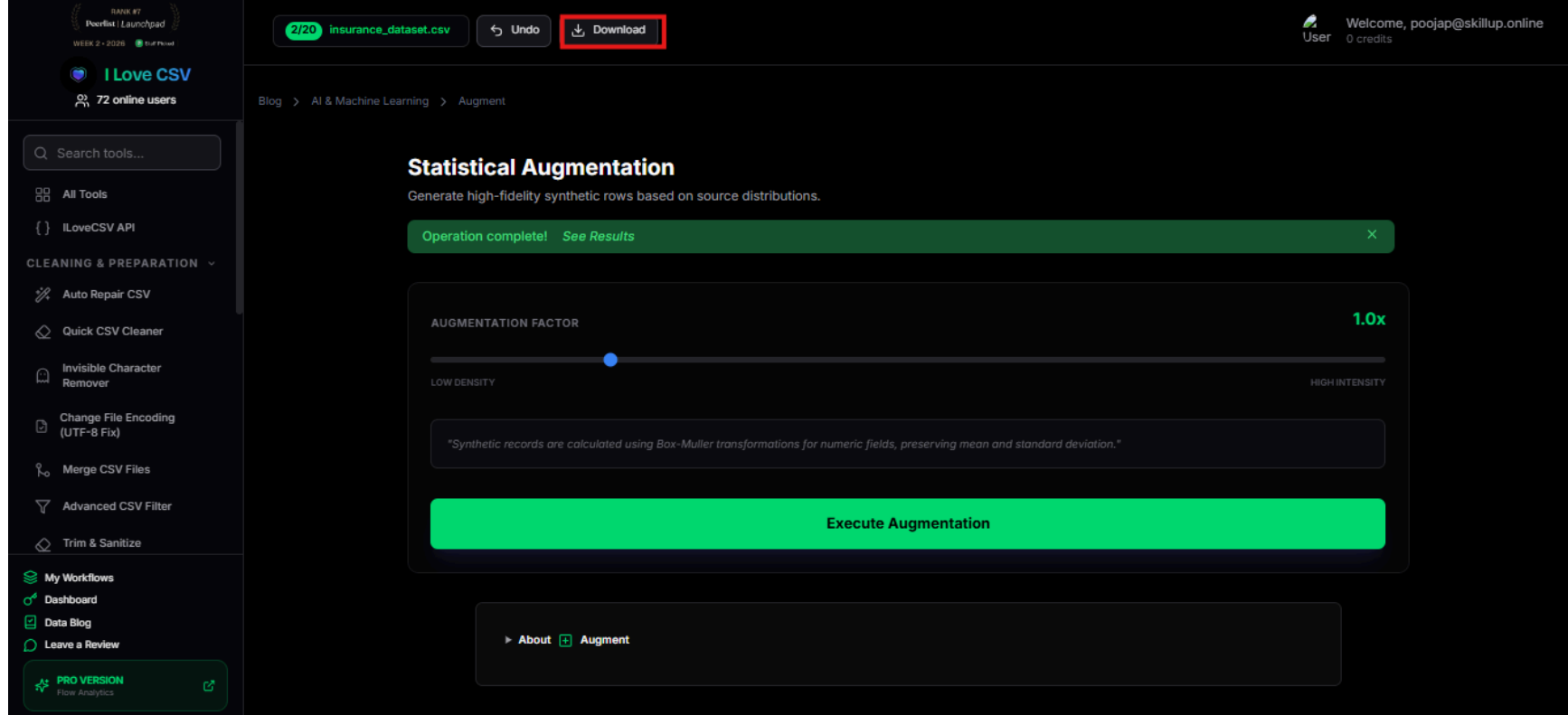
8

9

10

>

Close the results window and return to the previous page. Then click [Download](#) to save the output file to your local machine.



In the final result, you will get 2,676 rows, which is double the original file since you selected a 1× augmentation factor. You can now use this synthetic data set for data science operations, or you can also augment the original data set with these samples.

Option 2: Tonic Fabricate

1. Open the website

Click this [Tonic Fabricate](#) link to open the website and interface.

This link opens in a new browser tab, and you should see a web page that looks similar to the following screen capture:



TONIC FABRICATE

Generate a fake database in seconds.

 [How it works](#)

Create My Account



or use your email for registration

Name

Email

Password

6 characters minimum

Signing up signifies that you are authorized to enter into an agreement with Tonic, and that you have read and agree to our [Terms of Use](#) and [Privacy Policy](#), which will govern all uses of any Tonic product, unless there is a signed agreement between you and Tonic that you agree governs this use.

2. Create an account

You can create an account on this website free of charge, or you can simply log in using your Gmail ID. After you log in, you'll see the following interface:

What data can I fabricate for you today?

Drop in a schema file or describe the type of data you want to get started.

∞ Agent ▾



 Generate data for an online store

 Simulate bank transactions

 Create social media data

3. Data Augmentation

This is a prompt-based tool that lets you customize synthetic data to meet your data science operational requirements. Upload the Insurance dataset using the **Attach a File** option and copy and paste the below prompt in the message box.



plaintext



Generate a synthetic dataset based on the uploaded insurance data file. Preserve the original schema, column

Note: Synthetic data generation in this tool may take some time, so please be patient during the process.

What data can I fabricate for you today?

relationships between variables (e.g., higher age may correlate with higher premiums, and claim amounts should align with policy coverage). Avoid duplicates and unrealistic values. Generate clean, high-quality data suitable for analysis, keeping the synthetic data statistically consistent with the source dataset with approximately 80% similarity/accuracy. Create a dataset that is 1× the size of the original while introducing meaningful variations.

 insurance_dataset.csv (50.2 KB) 

∞ Agent ▾



 Generate data for an online store

 Simulate bank transactions

 Create social media data

Click **Send** to submit the prompt to initiate the process.

TONIC FABRICATE

Default

insurance

insurance_source

PLANS

WORKFLOWS

API

insurance_source

1338 records

Default

SQL

Memory

Import

Export

id	age	gender	bmi	children	smoker	region	expenses
INTEGER	INTEGER	TEXT	REAL	INTEGER	TEXT	TEXT	REAL
1	19	female	27.9	0	yes	southwest	16884.9
2	18	male	33.8	1	no	southeast	1725.55
3	28	male	33	3	no	southeast	4449.46
4	33	male	22.7	0	no	northwest	21984.4
5	32	male	28.9	0	no	northwest	3866.86
6	31	female	25.7	0	no	southeast	3756.62
7	46	female	33.4	1	no	southeast	8240.59
8	37	female	27.7	3	no	northwest	7281.51
9	37	male	29.8	2	no	northeast	6406.41
10	60	female	25.8	0	no	northwest	28923.1
11	25	male	26.2	0	no	northeast	2721.32
12	62	female	26.3	0	yes	southeast	27808.7
13	23	male	34.4	0	no	southwest	1826.84
14	56	female	39.8	0	no	southeast	11090.7
15	27	male	42.1	0	yes	southeast	39611.7

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Synthetic Insurance Data Generation

Children

0-5, heavily skewed toward 0 (43%)

Smoker

~80% no, ~20% yes

Region

~24-27% per region

Expenses

Strongly correlated with smoker status + age; smokers avg ~\$32K, non-smokers avg ~\$8.4K

Now I'll create the target table and generate 1,338 synthetic rows preserving these distributions and correlations:

Create synthetic insurance table

6s

Generate synthetic insurance data

24s

All 1,338 synthetic rows generated. Let me now validate the quality by comparing distributions between source and synthetic data:

Ran query source vs synthetic comparison

Ran query region distribution comparison

Ran query expense patterns by age & smoker

Working...

Agent

4. Download the synthetic data

Once the process is complete, you will be prompted with the message: **The data is ready for export or analysis. Would you like me to download it as a CSV?** Type **Yes** to proceed with downloading the dataset.

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Default

insurance

PLANS

WORKFLOWS

API

insurance

1338 records

SQL

Memory

Import

Export

id	age	gender	bmi	children	smoker	region	expenses
INTEGER	INTEGER	TEXT	REAL	INTEGER	TEXT	TEXT	REAL
1	33	male	29.2	3	no	southeast	5275.43
2	42	male	32.9	3	no	southeast	16588.7
3	43	female	38.6	2	no	southwest	4876.97
4	38	female	24.7	1	no	southwest	2868.68
5	62	female	28.3	1	no	northeast	14427.8
6	52	male	34	0	no	southwest	2065.15
7	63	female	39.9	0	no	southwest	23015.9
8	44	male	35.6	1	no	southwest	15470.2
9	53	male	35.7	0	yes	southeast	34589.7
10	24	male	26.2	3	no	northeast	4704.95
11	56	female	33	0	no	southeast	16720.5
12	24	female	27.8	1	no	southeast	1970.8
13	40	male	31.5	0	no	northeast	7378.22
14	47	male	33	0	yes	southwest	17645.3
15	49	male	31.2	1	no	southwest	11600.3

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Synthetic Insurance Data Generator

Yes

synthetic_insurance_dataset.csv

text/csv • 49.09 KB

Download

Here's your synthetic insurance dataset as a CSV (49 KB, 1,338 rows). The `id` column has been excluded to keep the format consistent with your original file.

What would you like to do next?

Agent

Click on **Download** to save the data to your local machine.

5. Analyze the report

This tool allows you to view details about the model used to generate the synthetic data.

Copy and paste the below prompt to see the report

<>

plaintext

Provide a detailed report on the model used for generating the synthetic data. Include information about the

After you give the prompt, you will see a detailed report along with visualizations.

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Default

insurance

PLANS

WORKFLOWS

API

insurance 1338 records

SQL

Memory

Import

Export

id	age	gender	bmi	children	smoker	region	expenses
INTEGER	INTEGER	TEXT	REAL	INTEGER	TEXT	TEXT	REAL
1	33	male	29.2	3	no	southeast	5275.43
2	42	male	32.9	3	no	southeast	16588.7
3	43	female	38.6	2	no	southwest	4876.97
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5	62	female	28.3	1	no	northeast	14427.8
6	52	male	34	0	no	southwest	2065.15
7	63	female	39.9	0	no	southwest	23015.9
8	44	male	35.6	1	no	southwest	15470.2
9	53	male	35.7	0	yes	southeast	34589.7
10	24	male	26.2	3	no	northeast	4704.95
11	56	female	33	0	no	southeast	16720.5
12	24	female	27.8	1	no	southeast	1970.8
13	40	male	31.5	0	no	northeast	7378.22
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15	49	male	31.2	1	no	southwest	11600.3

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Synthetic Insurance Data Generator

60-64	No	\$15,233	\$16,094	5.7%
60-64	Yes	\$40,631	\$42,178	3.8%

The **monotonic increase** in expenses with age is preserved for both smoker groups. Smokers consistently cost ~3-4x more than non-smokers across all age buckets.

8. Summary Scorecard

Evaluation Dimension	Weight	Score
Numeric Distribution (KS Tests)	30%	96.8%
Categorical Distribution (TVD)	20%	99.0%
Correlation Preservation	25%	96.3%
Discriminator AUC	25%	98.4%
Overall Similarity Score	100%	97.5%

The synthetic dataset exceeds the 80% similarity target across all dimensions, scoring 97.5% overall. The data is statistically consistent with the source, preserves all key relationships, and is indistinguishable from the original by a machine learning classifier.

What would you like to do next?

Agent

Conclusion

Congratulations! You have completed the lab on data augmentation using iLoveCSV and Tonic Fabricate tools.

Author(s)

Pooja Patel



